INNOMATİCS RESEARCH LABS

Text Data Preprocessing Assignment

1. Assignment Overview

This assignment is designed to provide hands-on experience with **Text Data Preprocessing**, one of the most important stages in Natural Language Processing (NLP) and Machine Learning workflows.

Students are expected to apply various preprocessing techniques to transform raw textual data into a clean and machine-learning-ready format.

2. Learning Objectives

By completing this assignment, students will be able to:

- Understand the importance of text preprocessing in Machine Learning.
- Perform common text cleaning operations.
- Apply tokenization techniques.
- Remove noise from textual data.
- Use stemming and lemmatization effectively.
- Create a final processed dataset suitable for Machine Learning models.
- Document and explain preprocessing decisions.

3. Assignment Tasks

Perform the following preprocessing steps on the provided text dataset:

1. IMDb Movie Reviews Dataset:
https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews

2. Amazon Product Reviews Dataset: https://www.kaggle.com/datasets?search=Amazon+Reviews

3. Restaurant Reviews Dataset: https://www.kaggle.com/datasets?search=Restaurant+Reviews

Select any one dataset from the sources listed above.

Ensure that the dataset contains a text or review column.

Task 1: Data Understanding

- Load the dataset.(collect the dataset from kaggle/any other authentic source)
- Frame the problem statement as per your perspective suitable for dataset
- Explore the dataset structure.
- Identify the text column(s) to be processed.
- Display sample records.

Task 2: Text Cleaning

- Convert text to lowercase.
- Remove punctuation marks.
- Remove special characters.
- Remove numerical values where appropriate.
- Remove extra whitespaces.

Task 3: Tokenization

- Split text into individual tokens (words).

Task 4: Stop Word Removal

- Remove commonly used stop words.

Task 5: Stemming

- Apply stemming and display sample outputs.

Task 6: Lemmatization

- Apply lemmatization and display sample outputs.

Task 7: Final Preprocessed Text

- Create a new column containing the fully preprocessed text.
- Compare original text and processed text samples.

Task 8: Summary

- Briefly explain:
    - What preprocessing techniques were applied.
    - Why each technique was used.

- Observations from the preprocessing process.

4. Deliverables

Students must submit the following:

Required Files

1. Jupyter Notebook (.ipynb)
2. Dataset used for the assignment
3. Processed Dataset (CSV file)
4. README or Summary Document (optional but recommended)

5. Notebook Structure

Students should organize their notebook using the following sections:

Section 1

- Assignment Title
- Student Name
- Batch Name

Section 2

- Import Required Libraries

Section 3

- Load Dataset

Section 4

- Data Understanding

Section 5

- Text Cleaning

Section 6

- Tokenization

Section 7

- Stop Word Removal

Section 8

- Stemming

Section 9

- Lemmatization

Section 10

- Final Preprocessed Dataset

Section 11

- Observations and Conclusion

6. Submission Guidelines

Step 1

Create a folder containing:

- Jupyter Notebook
- Dataset
- Processed Dataset
- Supporting files (if any)

Step 2

Compress the folder into a ZIP file.

Step 3

Upload the ZIP file to the LMS Assignment Submission Portal.

File Naming Convention

StudentName_TextDataPreprocessing.zip

Example:

JohnDoe_TextDataPreprocessing.zip

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7. Mandatory Mentoring Review

All students are required to attend a mentoring session and get their assignment reviewed.

During the Review

Students should be able to explain:

- Dataset overview
- Preprocessing workflow
- Tokenization process
- Stop word removal
- Stemming results
- Lemmatization results
- Final output

Assignment review during mentoring is mandatory and will be considered during evaluation.
